A common BSA–serum equation for EASI estimation in a Chinese atopic dermatitis cohort

Abstract

Background: Connecting serum inflammation with clinical severity requires a practical calculation that remains interpretable across treatment and follow-up. We evaluated a common Eczema Area and Severity Index (EASI) equation within a Chinese treat-to-target cohort.

Methods: We examined clinical outcomes and all 23 serum markers under dupilumab and two initial abrocitinib doses, adjusting treatment comparisons for measured baseline differences. A model-development procedure using nominal-visit-matched affected body surface area (BSA), CCL17, CCL26 and IL-5 was developed with baseline/week-12 observations using nested patient-level validation and applied at week 24. Individual estimates, within-person changes and EASI75 identification were evaluated separately, with matched input ablation and IgE missing-data sensitivity analyses.

Results: Clinical comparisons included 163 patients; EASI estimation included 316 visits from 160 patients. Both initial abrocitinib groups had lower adjusted early itch scores than dupilumab; some later differences persisted. Clinical control improved alongside treatment-dependent molecular changes. The selected four-input equation required no treatment label or previous score; performance estimates describe the nested development procedure. Mean absolute error (MAE) at baseline/weeks 12/24 was 3.19/1.30/1.19, with Spearman correlations of 0.918/0.927/0.904. Follow-up errors were within two points in 84.3% and 82.0% of predictions. Matched ablation attributed most predictive accuracy to BSA; adding current itch modestly improved the hybrid follow-up predictions. EASI75 sensitivity at weeks 12/24 was 76.5%/86.1%, with specificity of 98.6%/96.2%.

Conclusions: The BSA–serum development procedure gave close follow-up EASI estimates and tracked changes within this cohort. Molecular added accuracy was not established, and baseline severe disease remained the main accuracy boundary. These results support independent evaluation of this restricted-input research measure across the severity range.

Keywords: atopic dermatitis; biomarkers; EASI; longitudinal estimation; treat-to-target.

1 Introduction

Atopic dermatitis treatment must improve visible skin disease, itch and the burden of living with eczema. Treat-to-target frameworks bring these complementary outcomes into treatment decisions, including early symptom relief and later disease control.[1–3] A practical severity estimate should remain interpretable as treatment changes and should capture how an individual patient's condition changes, rather than only distinguish groups with different average severity.

Serum markers offer a link between clinical activity and inflammation. The p-EASI approach combines CCL17, IL-22 and soluble IL-2 receptor, with different coefficients before and during treatment.[4] Subsequent studies reported Spearman correlations of 0.71 during cyclosporin treatment and 0.67 during dupilumab treatment.[5, 6] These studies established a practical precedent for converting blood measurements into Eczema Area and Severity Index (EASI) units. Applying such a calculation in another setting also requires agreement with individual scores, compatible assays and evaluation across the severity range.[7, 8]

EASI combines the extent of affected skin with lesion intensity.[9, 10] A small-input equation combining affected body surface area (BSA) and serum measurements could offer a common calculation without separate rules for each drug or treatment phase. BSA supplies the clinical extent component; the serum panel supplies molecular information. We evaluated it as a research readout of recorded EASI at the visit, with separate tests of individual change and molecular added accuracy. Measured itch, quality of life and safety remained complementary clinical assessments.

We investigated these questions in a Chinese cohort receiving dupilumab or two initial abrocitinib doses. We first established the clinical and molecular context, retaining all 23 measured serum markers and accounting for baseline differences and later treatment changes. We then developed a common BSA–serum equation, evaluated individual estimates and changes using patient-level validation, and examined unchanged published p-EASI formulas as a separate local-application comparison.

2 Methods

2.1 Study population and clinical assessment

We analysed longitudinal records from a Chinese observational treat-to-target project at CAMS and Huashan. The supplied protocol specified Hanifin–Rajka diagnosis and eligibility for systemic treatment. Actual screening implementation and recruitment dates were not recoverable from a verified screening register. Systemic treatment was clinician directed. Initial regimens were dupilumab, abrocitinib 100 mg or abrocitinib 200 mg; later dose, interval and regimen changes were recorded separately. Clinical comparisons included patients aged at least 12 years with an unambiguous initial regimen. Adult Dermatology Life Quality Index (DLQI) analyses required age at least 18.

Clinical outcomes were the recorded EASI, peak pruritus numerical rating scale (PP-NRS), Investigator Global Assessment (IGA), DLQI, BSA and SCORAD, with IgE and eosinophil measurements. Reporting visits were W0/W2/W4/W12/W24 for PP-NRS, W0/W12/W24 for IgE and W0/W4/W12/W24 for other measures. Records were linked by patient and nominal visit. The supplied protocol describes a 24-hour PP-NRS recall window but its actual implementation, the exact IGA/questionnaire versions and calendar-day clinical/serum matching require confirmation. Source discrepancies were audited without substituting imputed clinical outcomes.

2.2 Clinical framework, targets and treatment pathways

Project materials describe evidence review, a two-round questionnaire procedure and a planned agreement criterion of at least 75%. They report 49 returned questionnaires, without original ballots or round-specific agreement results. We transcribed the September framework and operationalized its targets on recorded scores (Fig. 1). Its exact approved and prospectively implemented version remains unverified; this analysis does not validate a new consensus or the effectiveness of the management strategy.

The week-2 target was PP-NRS ≤ 4. At week 4, the physician target was EASI50 or an IGA decrease of at least two points, and the patient target was PP-NRS ≤ 4 or adult DLQI ≤ 10. At week 12, these were EASI75 or IGA 0/1, and PP-NRS 0/1 or adult DLQI ≤ 4. Week 24 additionally allowed EASI ≤ 5 in the physician domain. Both domains were required for the combined target. Individual reporting thresholds additionally included EASI ≤ 3; this did not change the stage-specific composite. These investigator-defined targets were kept distinct from newer low-disease-activity and remission definitions.[11]

Each rate used its own evaluable denominator and Wilson 95% confidence interval. EASI percentage responses required positive observed baseline EASI. A PP-NRS decrease of at least four points required observed baseline PP-NRS ≥ 4 and an observed follow-up score; lower or missing baseline values were excluded from that endpoint. Missing components remained unknown unless another observed component determined the logical result. Below age 18, adult DLQI was treated as unknown: a positive itch component establishes the patient target, whereas a negative itch component alone leaves it unknown. Group comparisons retained initial treatment after subsequent changes. The main pathway display summarized week-12 response and subsequent documented treatment changes as proportions of each complete initial group. Exploratory week-24 response summaries and missing-outcome bounds remain in the source tables. Reduction was not reassigned to an earlier visit; uncertain assessment/decision timing was retained. These descriptive pathways address treatment changes without treating a response-selected maintenance group as randomized.[12]

2.3 Serum measurements

Source laboratory reports described Luminex and MSD serum assays, including the LXSAHM-13 panel on Luminex-200 and K15067M-1/K15067M-2 kits on SQ120, using 50-μL serum and twofold dilution. All 23 measured markers were included. Analyses used positive numeric integration-export concentrations recorded in pg/mL, transformed to $\log_2$ once without an additional dilution correction. The twofold laboratory dilution describes sample preparation; its correction in the export and validated quantification limits require source confirmation. Censoring strings and nonpositive values were not treated as measured positive concentrations. An IL-5 calibration-bound sensitivity analysis and detailed assay provenance are provided in the Supplementary Methods; software calibration limits were distinguished from validated quantification limits.

Molecular changes were calculated within patient for W0–W12, W12–W24 and W0–W24 before summarization. Three-visit displays additionally held the patient set fixed within each marker and treatment. This separates individual change from shifts in the patients observed at each visit.

2.4 EASI estimation and validation

Eligible model visits required age at least 12, observed EASI and BSA, and positive observed CCL17, CCL26 and IL-5. This three-marker panel was fixed from earlier exploratory work. Development used W0/W12; W24 was excluded from fitting and selection. As W24 had been examined in previous project analyses, it represents within-cohort temporal application, not untouched external validation.

A repeated five-fold patient-level outer validation with three-fold inner validation selected among three common structures and ridge penalties. All records from a patient stayed in one outer fold, preprocessing used training data only, and patients received equal total fitting weight. The candidate structures were linear, quadratic in BSA, or an area interaction in which BSA multiplied a linear molecular term. Structure and ridge penalty were selected within the patient-grouped inner folds. The final coefficient vector was then fitted to all development observations; the equation is reported in Results.

Primary assessment used MAE with patient bootstrap intervals conditional on the saved held-out predictions; these intervals did not repeat model selection or fitting. This internal validation does not include the earlier selection of the marker panel. Additional measures were RMSE, signed error, Spearman correlation, Lin's concordance correlation and proportions within one or two EASI points. Repetition-specific metrics were calculated before averaging. Severity strata and centre/treatment holdouts assessed where performance varied. A separate refitting sensitivity removed development visits with complete regional scores whose sum differed from recorded total EASI by more than 0.15 points. It retained the original patient folds and outcomes and repeated inner selection; it tested exclusion robustness without correcting either clinical reference. Monitoring used estimated and observed changes within the same patient, repetition and fold; response classification used the corresponding estimated or observed scores at both ends of the interval. Matched controls included BSA alone and the three markers alone. Primary paired comparisons used the nested procedure; fixed-structure comparisons were labelled separately. Published p-EASI coefficients were applied unchanged as a source-table sensitivity analysis, using local IL2RA as an unbridged soluble IL-2 receptor proxy and retaining raw and matched-clipping results. Different inputs and unverified assay equivalence preclude a model-superiority claim.[4–6] PP-NRS, adult DLQI and recorded IGA were fitted independently with target-appropriate regression or ordinal models using the same restricted inputs.

Post hoc input ablation compared serum-only, BSA-only and hybrid models on identical visits and patient folds. A current-itch comparison retained 151/89/75 W0/W12/W24 observations; adult DLQI was tested separately on 105/58/44 matched adult observations. Every comparator was refitted within its comparison population. Structure and penalty selection remained inside the W0/W12 training folds. Fixed-area and single-marker controls distinguished molecular inclusion from changes in model structure. Paired patient bootstrap differences were exploratory and not adjusted across the multiple input comparisons; full specifications appear in the Supplementary Methods.

2.5 Statistical analysis

Patients were the independent units. Baseline continuous variables were summarized by the median and interquartile range, using linear interpolation at sorted rank $1+(n-1)p$. This matches Excel PERCENTILE.INC; changing the quartile convention changes summaries without changing patient records. Main treatment-specific trajectories show observed medians and interquartile ranges; adjusted mean comparisons are reported separately. Continuous treatment comparisons used baseline-outcome adjustment, treatment-by-visit terms, age, sex, BMI and centre, generalized overlap weights and patient-cluster robust covariance. Weights target overlap in baseline treatment probabilities. Continuous-outcome intervals condition on the estimated baseline weights; they do not propagate propensity-estimation uncertainty. Reported outcome contrasts are baseline-weighted associations among patients with that endpoint observed. Selective endpoint observation can therefore change the represented population and its covariate balance.[13] Weighted target contrasts used patient bootstrap intervals with propensity refitting. A W2 sensitivity additionally included baseline PP-NRS, comparing models on the same baseline-itch-complete cohort. Endpoint-specific balance, effective sample sizes and both weighted and unweighted missing-outcome bounds were retained; weighting did not correct informative follow-up loss. IgE/eosinophil models used $\log(1+x)$ and back-transformation. The analyses estimate adjusted observational associations.

IgE observation-pattern sensitivity used all available within-patient pairs and a maximum-likelihood random-intercept model. Patient-wide multiple imputation compared within-treatment predictive mean matching with posterior log-normal draws, retaining 400 datasets per method. CAMS was the primary sensitivity population because every treatment had observed follow-up there; full-cohort imputation required additional cross-centre assumptions. Observed values remained fixed, and no imputed clinical score entered prediction validation. MAR and illustrative half/double missing-value scenarios were kept separate from measured trajectories. Rubin pooling, chain diagnostics and donor-support checks are detailed in the Supplementary Methods.[14, 15]

Unadjusted target comparisons used conditional Monte Carlo three-group tests and pairwise Fisher tests; the three pairwise contrasts within each endpoint/visit formed a Holm family. Molecular paired sign-flip tests used

Benjamini–Hochberg adjustment over 23 markers, three treatments and three windows. Between-treatment molecular changes additionally adjusted for starting concentration and clinical baseline variables. Estimates based on fewer than five pairs were descriptive. Balance, effective sample size and common support were examined. Exact tests, numerical settings, model diagnostics, regional analyses and additional targets are detailed in the Supplementary Methods. Analyses used Python 3.9.6; dependency versions, code, data-access conditions and reproduction logs accompany the package. Reporting follows relevant STROBE and TRIPOD+AI domains.[16, 17]

3 Results

3.1 Clinical framework and analysis populations

The source framework combined physician and patient criteria at W4, W12 and W24; W2 assessed itch alone (Fig. 1). The serum-assisted EASI estimate was evaluated as a parallel research measure and did not determine dose-reduction decisions.

The clinical comparison included 163 patients: 65 started dupilumab, 56 abrocitinib 100 mg and 42 abrocitinib 200 mg. Median baseline EASI was 15.20, 11.70 and 19.35, respectively; age, sex and BMI also differed (Table 1). Clinical treatment comparisons and linked-serum modelling used separate eligibility rules, detailed in Supplementary Fig. S1.

Table 1: Baseline characteristics of the clinical comparison population.

	Dupilumab (n=65)	Abrocitinib 100 mg (n=56)	Abrocitinib 200 mg (n=42)
Age, years	54.0 (29.0–71.0)	37.5 (22.0–52.0)	24.5 (19.0–46.8)
BMI, kg/m^2	23.0 (21.0–26.0)	22.4 (20.2–24.3)	23.6 (21.4–26.0)
Male sex, n/N (%)	36/65 (55.4)	27/56 (48.2)	35/42 (83.3)
EASI	15.2 (11.9–21.6)	11.7 (7.7–17.3)	19.4 (9.6–27.8)
PP-NRS	8.0 (7.0–9.0)	7.0 (5.5–8.0)	7.0 (5.0–9.8)
Available n	65	55	42
IGA, recorded grade	4.0 (3.0–4.0)	3.0 (3.0–4.0)	4.0 (3.2–4.8)
Adult DLQI	15.0 (10.5–21.0)	10.5 (7.0–15.2)	15.0 (10.2–21.8)
Available n	47	32	34
BSA, %	23.0 (17.0–37.0)	16.0 (10.0–25.2)	25.0 (15.5–36.2)
SCORAD	50.0 (43.2–59.4)	48.1 (40.4–54.2)	58.1 (48.7–72.6)
Available n	65	44	41
Eosinophils, recorded units	0.2 (0.1–0.4)	0.3 (0.1–0.6)	0.5 (0.3–0.7)
Available n	34	51	38
IgE, recorded units	200.0 (57.2–741.2)	214.7 (71.0–626.0)	700.0 (57.6–5000.0)
Available n	62	41	37

Values are median (interquartile range), unless stated otherwise. Quartiles use linear interpolation at rank $1 + (n - 1)p$ in the sorted observations (equivalent to Excel PERCENTILE.INC). Adult DLQI is restricted to age ≥ 18. IGA is retained on the recorded scale pending instrument confirmation. The column-header n applies unless an available n is shown. Missing values are not filled.

3.2 Clinical control improved under the three initial treatment strategies

Observed EASI, PP-NRS and adult DLQI decreased from W0 to W12 and W24; the component target rates describe the corresponding clinical control (Fig. 2). At W2, PP-NRS was at most 4 in 15/27 evaluable dupilumab recipients, 34/37 receiving initial abrocitinib 100 mg and 33/36 receiving initial abrocitinib 200 mg. The adjusted 100 mg–dupilumab difference was 41.7 percentage points (95% CI, 15.1–69.4; Holm-adjusted $P = 0.0095$). Adding baseline itch gave a similar difference of 41.4 points (13.0–66.8). W2 itch was unavailable in 38/65, 19/56 and 6/42 patients, respectively, leaving these associations sensitive to follow-up selection. Weighted worst-case bounds permitted reversal of the W2 contrast (Supplementary Methods).

Continuous itch comparisons also favoured both initial abrocitinib groups at W2. Adjusted PP-NRS was lower than with dupilumab by 2.14 points for 100 mg (95% CI, 0.82–3.46) and 2.24 for 200 mg (0.83–3.65; both Holm-adjusted $P = 0.0049$). Differences remained at W24: dupilumab-minus-abrocitinib contrasts were 1.62/1.60 PP-NRS points and 2.98/3.32 EASI points for 100/200 mg, respectively, with all four Holm-adjusted $P < 0.015$. The estimated itch gaps were 0.52 and 0.64 points smaller at W24 than W2, but direct intervals for these changes included zero (-1.98 to 0.93 and -2.31 to 1.03 for later-minus-early differences). These data support an early itch association, without establishing convergence or equivalence of later efficacy. Complete estimates and intervals are supplied in the clinical contrast tables.

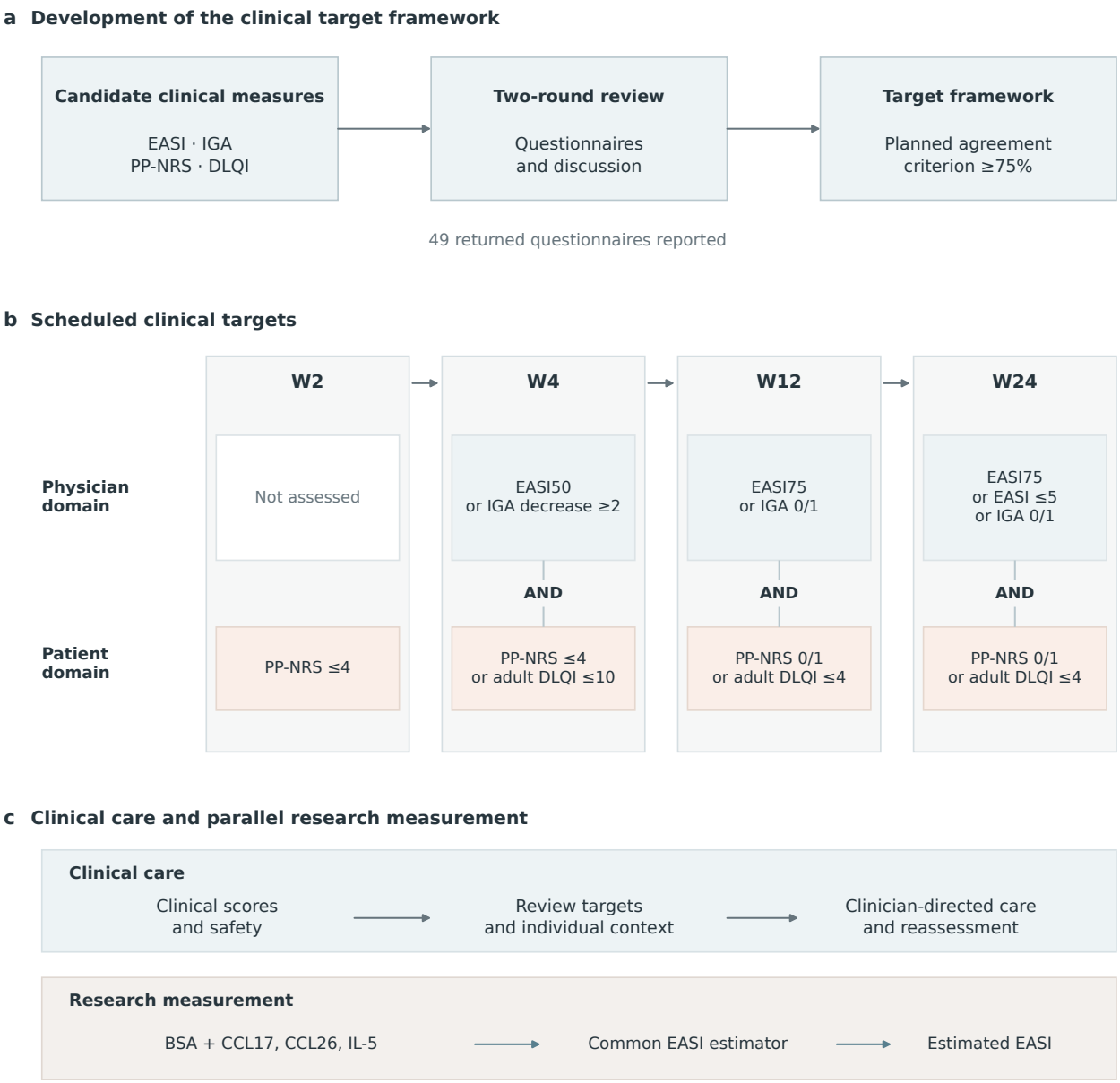
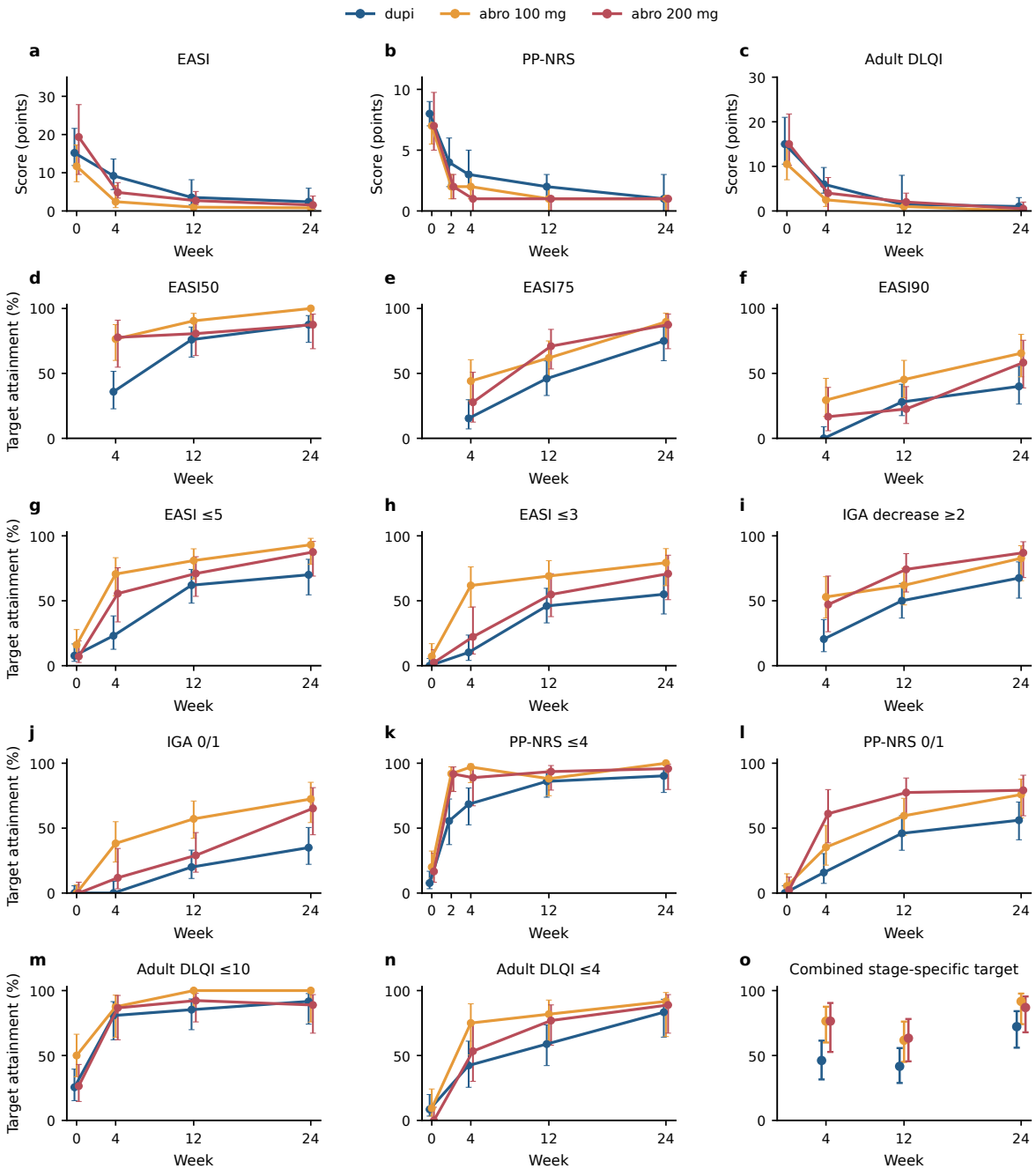


Figure 1: **Clinical target framework and parallel research EASI estimation.** a, Framework development described in the investigator materials: candidate clinical measures, a two-round questionnaire and discussion procedure, 49 returned questionnaires, and a planned agreement criterion of at least 75%. Original ballots and round-specific agreement were unavailable; these are source-reported development details, not independently verified consensus results. b, Scheduled targets transcribed from the project framework. Within each domain, any listed criterion is sufficient; from W4 onward both domains must be met. W2 assesses only PP-NRS $\leq$4. EASI50 and EASI75 denote at least 50% and 75% improvement from a positive observed baseline EASI. IGA decrease $\geq$2 is relative to baseline; IGA 0/1 and PP-NRS 0/1 denote observed values 0 or 1. DLQI targets apply to adults aged $\geq$18 years. Missing components are handled with three-valued logic, rather than automatically classified as target failure. The exact IGA instrument and approved implemented framework version require investigator confirmation. Visits are schematic and not spaced in proportion to time. c, Clinician-directed assessment and care are separated from the research estimator using BSA, CCL17, CCL26 and IL-5. The estimator was not a dose-change rule, and this observational analysis does not establish the effectiveness of the proposed integrated management strategy.



Evaluable n / initial N			**k · PP-NRS ≤4**			**o · Combined target**		
	W0	W2	W4	W12	W24	W4	W12	W24
dupi	65/65	27/65	38/65	50/65	41/65	39/65	48/65	36/65
abro 100 mg	55/56	37/56	34/56	42/56	29/56	34/56	34/56	24/56
abro 200 mg	42/42	36/42	18/42	31/42	24/42	17/42	30/42	23/42

Figure 2: **Observed clinical activity and target attainment by initial treatment.** a–c, Observed medians and interquartile ranges for EASI, PP-NRS and adult DLQI. d–n, Clinical threshold attainment, with pointwise Wilson 95% confidence intervals. Each visit uses its available observations, not a fixed patient set. Group positions are slightly offset for legibility. Improvement thresholds require observed baseline and follow-up; adult DLQI requires age ≥18. o, Combined physician-and-patient targets from Fig. 1, with Wilson intervals. These stage-specific criteria differ, so points are not connected. Colour denotes initial treatment throughout; subsequent treatment changes do not reassign groups. The footer gives evaluable n over the initial-group N for k and o. No outcomes are imputed. All target numerators and denominators are in Supplementary Table S1; headline missing-outcome bounds are in Results and headline_missing_outcome_bounds.csv. Baseline-adjusted treatment comparisons are in the supplement. dupi denotes dupilumab; abro denotes abrocitinib. These available-case, observational trajectories do not establish causal treatment differences.

Recorded IGA distributions shifted towards lower grades, while later management included both continuing and reducing treatment (Fig. 3). At W24, the combined physician- and patient-reported target was reached by 26/36 (72.2%), 22/24 (91.7%) and 20/23 (87.0%), respectively (Fig. 2o). Counting all unknown outcomes as failures or successes gave initial-cohort bounds of 40.0–84.6%, 39.3–96.4% and 47.6–92.9%. The pathway proportions retain all initial patients and separate W12 response status, subsequent treatment reduction and unavailable management records. Their composition describes management after response; it does not estimate a randomized dose-reduction effect.

Removing the IGA component from the reconstructed framework changed combined-target success to failure in 4/90, 6/112 and 0/83 jointly evaluable patients at W4/W12/W24, respectively (Supplementary Table S10). This definition sensitivity preserved the W24 combined classifications; it did not validate the implemented IGA scale.

The lower unadjusted EASI in the 100 mg group did not persist as a clear dose-group difference after baseline adjustment. The adjusted 100-minus-200 mg difference was 0.62 points at W12 (95% CI, −2.08 to 3.33) and 0.34 at W24 (−1.10 to 1.77). Weighting improved baseline balance, although endpoint-specific residual imbalance remained (Supplementary Fig. S2). Three W0 regimen entries conflicted with the initial-dose label and remain unresolved. Safety documentation through W24 was present for 27/65, 20/56 and 19/42 patients; these counts do not establish event incidence or treatment attribution (Supplementary Table S3).

BSA, SCORAD and laboratory measures appear in Supplementary Figs. S3–S4; all target contrasts appear in Supplementary Figs. S6–S9. No observed outcome was replaced. IgE follow-up selected different patients: in the abro 200 mg group, baseline IgE was 3303 among patients with W24 IgE, versus 174 among those without it (medians in recorded units). Using all available W0/W24 pairs across both centres retained 14/12/13 patients across the three treatments, compared with 11/9/10 complete three-visit patients. Within CAMS, available-data modelling and two imputation methods supported lower IgE with dupi, but the W24 direction for abro 200 mg depended on the missing-data model and remained uncertain (Supplementary Fig. S5; Supplementary Table S8).

Eosinophil coverage was also uneven: dupi had no W4 measurements, and only 10 dupi patients had both W0 and W24 values. Their mean paired change was −0.158 recorded units (95% CI, −0.405 to 0.089). Corresponding changes were −0.089 (−0.208 to 0.030; 23 pairs) for abro 100 mg and −0.263 (−0.442 to −0.084; 22 pairs) for abro 200 mg. These descriptive within-person changes do not establish a difference between treatments. Coverage and all three paired intervals are provided in Supplementary Table S9.

3.3 The complete serum panel revealed treatment-dependent early changes

All 23 measured markers were retained, including CCL18, IL-13, IL-22 and IL-31. Figure 4 emphasizes W0–W12, separating each treatment's paired changes from baseline-adjusted differences between treatments. Within-group significance alone does not establish a treatment difference. The dose comparison retains both abrocitinib groups; the dupilumab-versus-pooled-abrocitinib estimate represents the observed mixture of the two doses.

For CCL17 over W0–W12, the adjusted dupilumab-minus-abro 100 mg difference was −1.07 $\log_2$ units (95% CI, −1.67 to −0.48; q=0.0109). Its assay-batch sensitivity estimate was −0.72, with an interval crossing zero (q=0.369). This contrast therefore depended on the adjustment specification. The baseline distributions, later windows and remaining adjusted comparisons are retained in Supplementary Figs. S10, S11–S15. The three-visit display fixes the patient set within each marker and treatment (Supplementary Fig. S18); the window analyses use every eligible pair.

Selected molecular changes accompanied clinical changes. In abro 100 mg over W0–W12, IL-5 change and PP-NRS change had Spearman $\rho = 0.742$ (31 pairs; panel-wide q=0.00156). The declared association family also identified TNF-α and TSLP changes with EASI change over W0–W24. Supplementary Fig. S16 displays all associations meeting the existing panel-wide FDR criterion, across all windows. The complete family, including nonsignificant and non-estimable cells, remains in the source tables; selection for this display is exploratory. Regional disease and baseline molecular associations are brought together in Supplementary Fig. S17.

3.4 One equation estimated follow-up EASI closely across visits

The model dataset contained 316 complete-input visits from 160 patients: 152 at W0, 89 at W12 and 75 at W24. Development used 241 W0/W12 visits from 156 patients. The final equation combined BSA, CCL17, CCL26 and IL-5 with the same calculation before and during treatment. It required no treatment label, visit week or previous score. Observed-versus-estimated score distributions are shown in Supplementary Fig. S19.

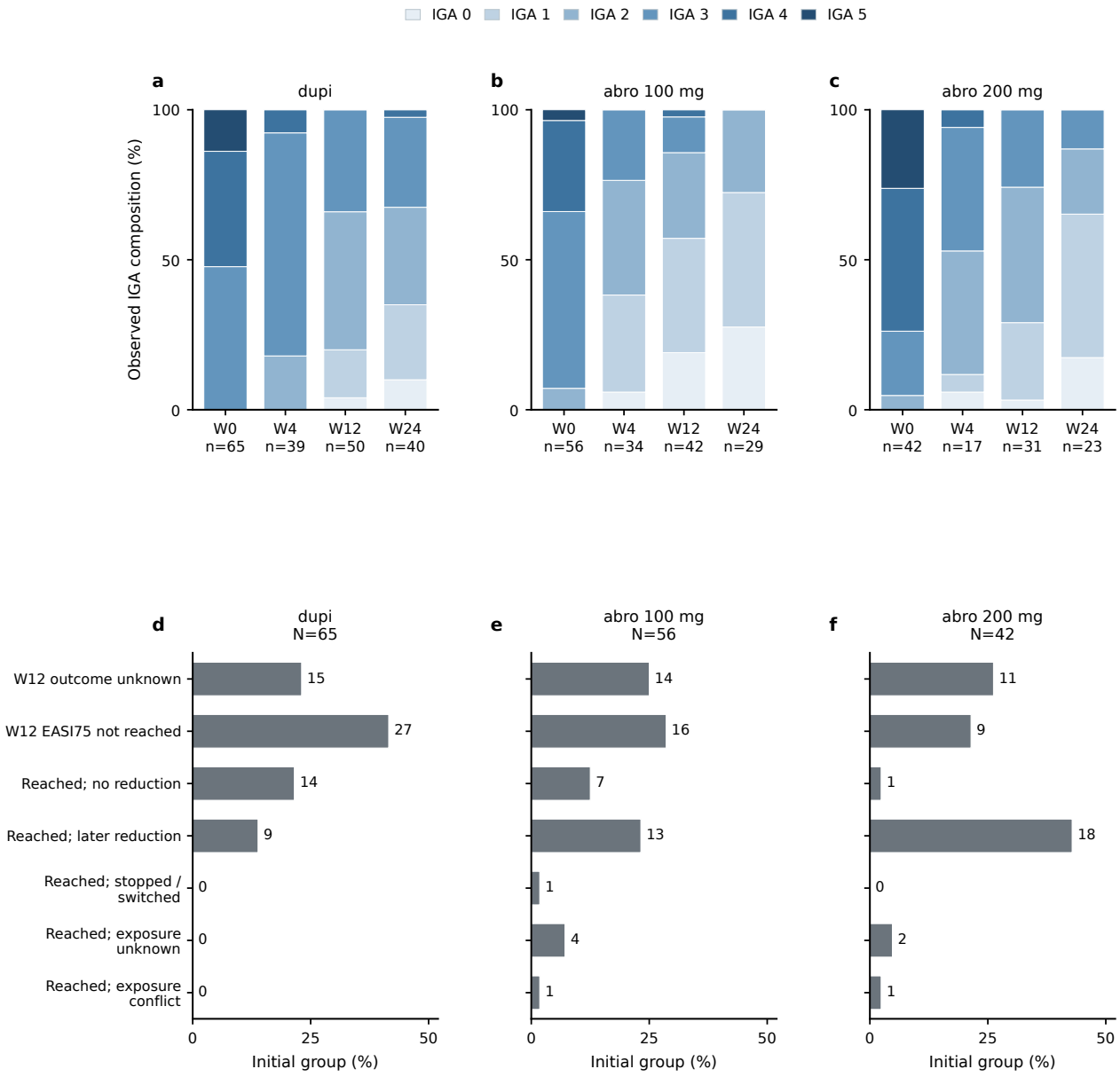


Figure 3: **Recorded skin severity and treatment-reduction pathways.** a–c, Recorded IGA compositions at W0, W4, W12 and W24. Each bar totals 100% of the evaluable patients; n below the bar is its patient denominator. The ordered blue scale identifies recorded IGA grades 0–5 and is not a treatment colour. No score or count is superimposed on the segments. All grade-specific counts and proportions remain in source tables. d–f, Future-defined pathways formed by W12 EASI75 status and subsequent recorded treatment changes. Horizontal bar length is the proportion of the complete initial group (N=65/56/42), and the number at its end is the pathway size. The seven mutually exclusive rows exhaust each initial group, including zero-size and uncertain-exposure rows. Reached refers to W12 EASI75; later reduction means a documented reduction after W12. These are descriptive pathways, not randomized reduction strategies; the bars do not encode subsequent efficacy or relapse. Instrument identity remains unverified for the recorded IGA scale. Complete pathway counts are available in source data.

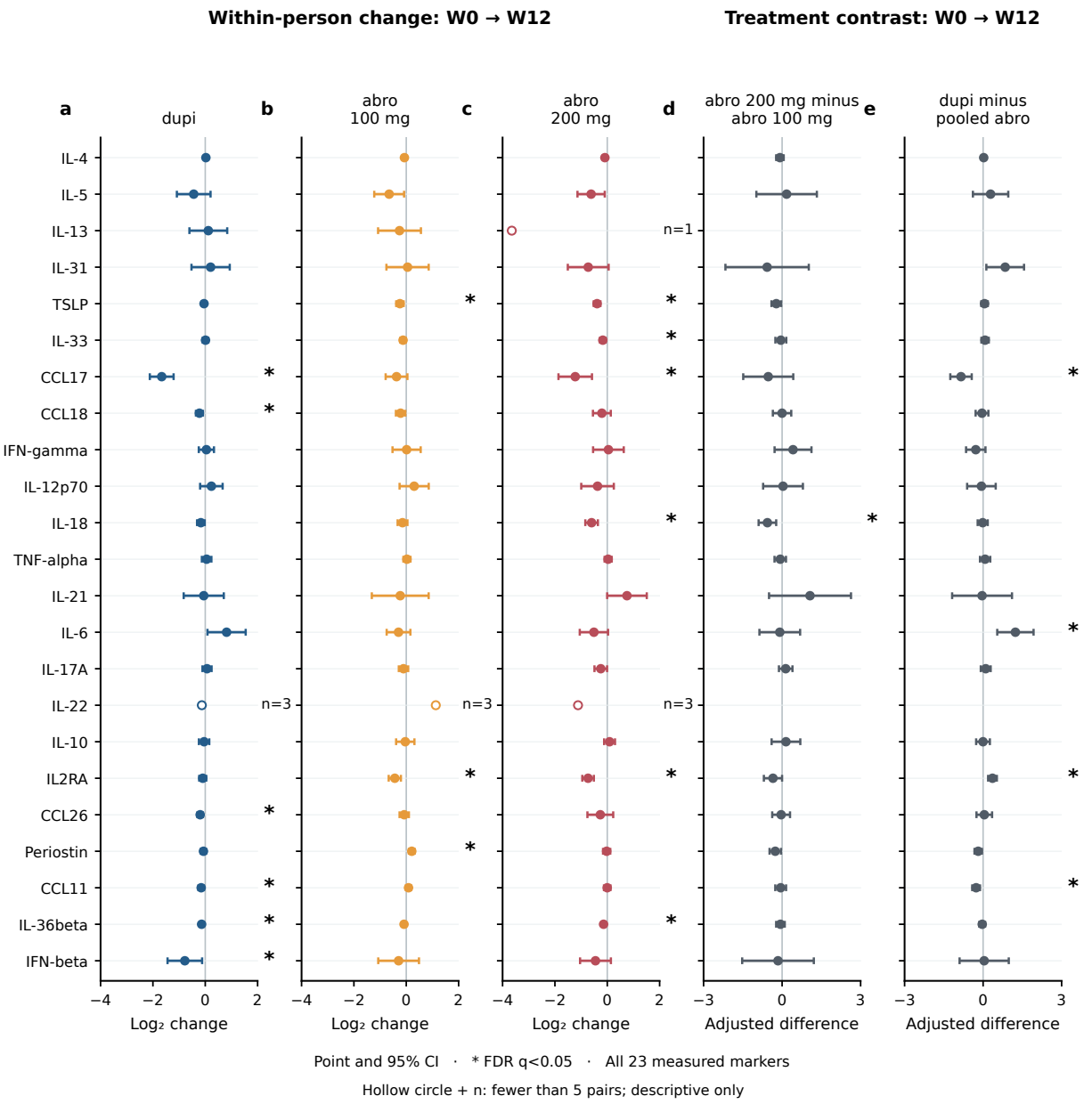


Figure 4: **Serum molecular response during the first 12 weeks.** a–c, Mean within-patient change in $\log_2$ concentration from W0 to W12 and 95% t confidence intervals for all 23 markers, by initial treatment. A change of −1 corresponds to half the baseline concentration. d–e, Direct adjusted differences in those changes for abro 200 mg minus abro 100 mg and dupi minus pooled abro, with HC3-robust 95% confidence intervals. Models adjust for starting $\log_2$ concentration, W0 EASI, age, sex, BMI and centre. pooled abro preserves the observed dose mixture. Negative contrasts mean a larger decline in the first group. Stars show q<0.05 under the original declared BH families; no tests or multiplicity families were changed for this display. Each marker uses its own available pairs. Hollow circles with n labels denote fewer than five pairs and have no inferential interval or star; missing estimates are left blank. Exact sample sizes, estimates, intervals and q values remain in source tables. Within-group and between-group panels have separately labelled x axes and cannot be interpreted as the same effect. The remaining pairwise treatment contrasts, later windows, fixed three-visit trajectories and assay-batch sensitivities are reported in the supplement. Treatment assignment was observational.

The fitted equation multiplied affected area by a serum-based component. The two-decimal display was

$$\widehat{\text{EASI}} \approx \text{clip}_{[0,72]}\Big\{\frac{\text{BSA}}{100}\big[51.27 + 0.98\log_2(\text{CCL17}) + 0.96\log_2(\text{CCL26}) - 0.83\log_2(\text{IL-5})\big]\Big\}. \quad (1)$$

BSA is entered from 0 to 100 and concentrations use the development export's recorded pg/mL numerics, with the logarithm applied once. Here, clip bounds the result to 0–72, and $\log_2$ is the base-two logarithm. The project label Skin inflammation score refers here to this hybrid EASI estimate, not a separately calibrated blood-only inflammation scale. The calculator retains full-precision parameters; the displayed approximation differed from final-equation predictions by less than 0.08 EASI points in this dataset. Threshold decisions use the precise calculator. Treatment, stage, time and previous scores are not inputs. Reported primary performance evaluates the nested selection procedure; it is not the apparent fit of this final equation.

MAE under nested patient-level validation was 3.19 at W0 (95% CI, 2.75–3.70) and 1.30 at W12 (0.96–1.71); same-cohort W24 temporal application gave 1.19 (0.89–1.53). Spearman correlations were 0.918, 0.927 and 0.904; concordance correlations were 0.910, 0.900 and 0.869 (Table 2). The error distributions (Fig. 5d) showed that 41.8%, 84.3% and 82.0% of predictions were within two EASI points. These describe individual errors, rather than agreement between group means. Mean signed errors at W12/W24 were +0.82/+0.84 points, with calibration slopes of 0.81 (95% CI, 0.66–0.94) and 0.73 (0.59–0.82), indicating residual overestimation and scale miscalibration.

Table 2: Primary performance of the common-model development procedure and matched controls.

	W0	W12	W24
Patients	152	89	75
MAE (95% CI)	3.19 (2.75, 3.70)	1.30 (0.96, 1.71)	1.19 (0.89, 1.53)
BSA-only area MAE	3.12	1.18	1.02
Three-marker-only MAE	7.63	7.57	8.69
Primary minus BSA-only area			
MAE difference (95% CI)	0.07 (−0.03, 0.17)	0.12 (0.04, 0.20)	0.17 (0.11, 0.24)
RMSE	4.45	2.29	1.87
Mean signed error	−0.59 (−1.27, 0.07)	0.82 (0.43, 1.29)	0.84 (0.50, 1.25)
Spearman correlation	0.918	0.927	0.904
Concordance correlation	0.910	0.900	0.869
Within two points, %	41.8	84.3	82.0
Severity-band agreement, %	83.2	88.2	92.7

W0/W12 use patient-grouped nested validation; W24 is same-cohort temporal application. Structure and penalty selection are internal to each training fold; earlier marker-panel selection is not repeated. Matched controls use the same patients and outer folds. BSA-only area uses the zero-intercept area structure; three-marker-only uses CCL17, CCL26 and IL-5 without BSA. Positive paired MAE differences mean greater primary-procedure error. Metrics are averaged after scoring each repetition. Intervals are 95% patient-bootstrap intervals conditional on saved predictions; selection and fitting are not repeated. MAE, mean absolute error; RMSE, root mean squared error. Positive signed error indicates overestimation. Severity-band agreement uses four EASI bands for observed and predicted values: ≤ 7, > 7 to ≤ 21, > 21 to ≤ 50, and > 50. The single final formula is not independently validated.

Baseline severe disease remained the main accuracy boundary: MAE was 5.38 among 44 patients with EASI > 21, compared with 1.15 and 2.55 in the lower strata. Full severity-stratified estimates remain in the model source tables. Only one W12 observation and no W24 observation had EASI > 21. Centre and treatment holdouts also varied; their complete estimates remain in the model source tables.

Removing eight W0 and two W12 reference-inconsistent visits from training, while retaining the original folds and evaluation outcomes, gave sensitivity MAEs of 3.19/1.15/1.06. This improved follow-up error but did not establish molecular added accuracy over the identically refitted BSA-area control. The frozen primary equation and primary estimates were retained (Supplementary Table S4). Independent restricted-panel models for PP-NRS, adult DLQI and recorded ordinal IGA were less accurate. W12/W24 MAEs were 2.78/2.53 for PP-NRS and 5.90/6.28 for DLQI; exact IGA agreement was 35.4%/28.0% (Supplementary Methods).

3.5 Area and symptom inputs accounted for most predictive accuracy

Matched input ablation used the same 151/89/75 observations at W0/W12/W24 for every comparison (Fig. 5a–c). Adding BSA to the three serum markers reduced MAE from 7.67/7.48/8.56 to 3.20/1.29/1.19 points. Adding current PP-NRS further reduced follow-up error to 1.17 and 1.04 points; paired MAE differences were −0.127 (95% CI, −0.236 to −0.011) and −0.145 (−0.246 to −0.047). The matched BSA-plus-PP-NRS controls gave 1.10 and 0.96 points, so this improvement was attributable to symptom information rather than an established serum advantage.

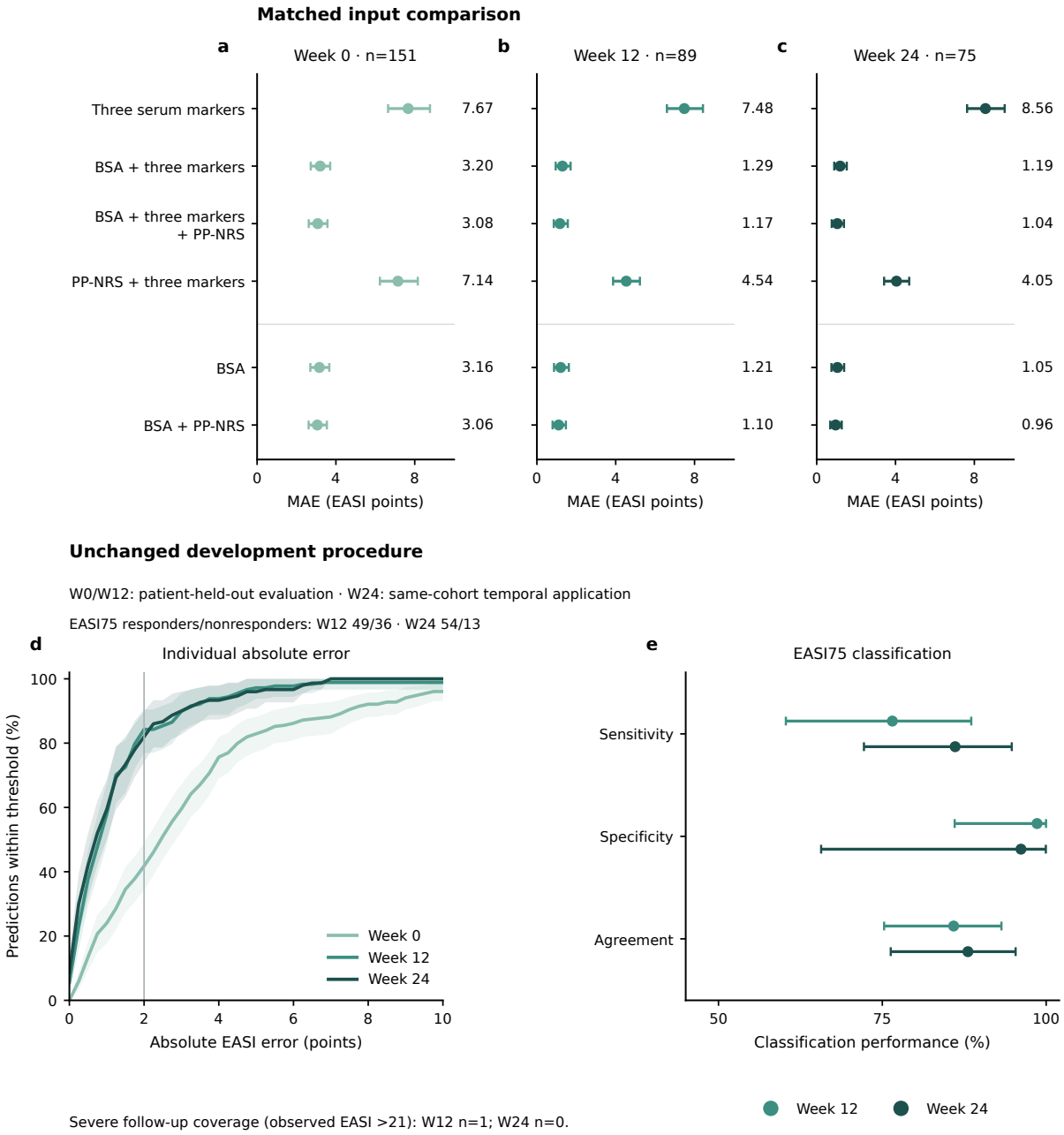


Figure 5: **Input contributions and performance of the common EASI estimator.** a–c, Post hoc input-set comparisons on the same 151/89/75 patients at W0/W12/W24. Points and labels show mean absolute error (MAE) with 95% paired-patient bootstrap intervals. Markers are CCL17, CCL26 and IL-5; clinical inputs come from the current visit. Models use the same outer patient folds, W0/W12 training, and training-only preprocessing and inner selection. Clinical-only controls appear below the divider; molecular added accuracy requires comparison at identical clinical inputs. d, The unchanged primary procedure on its original 152/89/75 patients: cumulative absolute-error proportions with pointwise bootstrap 95% intervals. All errors enter the calculation; thresholds up to ten points are displayed. Two points is a visual reference, not an accepted interchangeability margin. e, EASI75 classification uses observed and estimated scores with their own baselines. Sensitivity uses 49/54 responders, specificity 36/13 nonresponders and agreement 85/67 paired patients at W12/W24. Conservative conditional 95% bounds average repetition-specific Clopper–Pearson limits with Bonferroni adjustment across repetitions within each endpoint, visit and metric, not all displayed metrics. The binomial working model permits dependent repetitions but does not propagate shared fitted models or development uncertainty. Metrics are scored per repetition before averaging; intervals condition on saved predictions. W24 is previously inspected same-cohort application. Separate Wilson intervals and counts are in Supplementary Table S11; additional input comparisons are in Table S6.

Three markers did not lower MAE relative to CCL17 alone. With the area structure held constant, adding three markers to BSA reduced follow-up MAE by only 0.031 and 0.015 points, with both intervals spanning zero. In the separate adult DLQI subset, adding the serum panel to BSA, PP-NRS and DLQI reduced W24 MAE by 0.072 points (44 patients; 95% CI, 0.021–0.124), while increasing baseline error. These exploratory input comparisons did not establish that all three markers were necessary or optimal (Supplementary Tables S6–S7). The frozen equation was retained; the expanded-input models were not substituted for it.

3.6 Monitoring precision depended on the interval being assessed

Estimated changes tracked the ordering of observed improvement, with different absolute errors across intervals (Supplementary Fig. S20). Change MAEs were 3.87 for W0–W12 (85 patients; 95% CI, 3.08–4.70), 3.47 for W0–W24 (67 patients; 2.68–4.39) and 1.49 for W12–W24 (55 patients; 1.00–2.19). Corresponding Spearman correlations were 0.873, 0.940 and 0.900. The two intervals starting at baseline retained greater error despite their high correlations.

Using estimated scores at both visits to identify EASI75 (Fig. 5e) gave sensitivities of 76.5% at W12 and 86.1% at W24, with conservative conditional 95% bounds of 60.3–88.6% and 72.2–94.8%. Specificities were 98.6% (86.0–100%) and 96.2% (65.7–100%), based on 36 and 13 observed nonresponders. The broad bounds reflect the limited nonresponder samples; they combine repetition-specific binomial intervals, not independent copies of the patients. Sensitivity for EASI90 was lower, at 63.0% and 52.9% (Supplementary Fig. S21). These paired evaluations support examining response alongside the absolute reading, while preserving the distinction between monitoring agreement and clinical interchangeability.

4 Discussion

A common four-input model-development procedure connected nominal-visit-matched affected skin area and serum measurements to EASI throughout this Chinese treatment cohort. Its strongest performance was in follow-up disease activity, where most estimates were within two EASI points. Using the same calculation across visits also allowed changes to be expressed directly in EASI units. This makes the formula straightforward to examine and reproduce: the research equation takes BSA and three serum concentrations without a treatment label or previous score. Same-day clinical/serum matching was not verified, so its use as a contemporaneous clinical measurement requires a defined sampling window.

The equation combines visible disease extent with measured serum information. BSA is an established extent component of EASI, which explains why it supplied most predictive accuracy in the ablation.[9, 10] CCL17, CCL26 and IL-5 provide biological measurements, but including laboratory inputs does not make the hybrid score observer-independent. The matched input comparison did not establish that these assays are needed to estimate EASI. Their treatment-dependent trajectories answer a separate biological question and should not be used as evidence of predictive benefit. CCL17 and CCL26 declined in selected treatment comparisons, while IL-5 change accompanied itch change in one group. These observations address inflammatory response; they do not prove that three assays improve EASI estimation. Skin-biomarker findings during abrocitinib similarly show that molecular response and clinical correlation need not coincide, although tissue expression and serum concentrations are different measurements.[18] Conditional coefficient signs, including the negative IL-5 term, are not mechanistic weights. A separate blood-only inflammation scale or a claim of molecular necessity would require evidence beyond the present fitted EASI reference.

The expanded-input comparisons offer a practical trade-off. Current itch modestly improved the hybrid follow-up estimate, and adult DLQI added some information in a smaller subset, at the cost of additional subjective inputs. Their performance does not justify relabelling the result as a purely objective score. The appropriate choice depends on whether the intended output is a short clinical estimate or an integrated clinical–serum assessment; the current data support the latter as an investigational measurement, with molecular added accuracy still unestablished across visits.

Individual monitoring is a distinct test from agreement at a single visit. The formula reproduced much of the ordering of clinical improvement, yet changes beginning at baseline were less precise than later follow-up changes. Higher observed baseline severity was associated with larger errors. Severity mix, prior treatment and reference quality cannot be separated as causes by these data. Lesion intensity can vary despite similar affected area. Chinese serum endotypes and recent severity-linked proteomic findings also show broader inflammatory and epithelial-associated heterogeneity than a three-marker panel can represent.[19, 20] These observations motivate testing additional lesion information in severe disease, but they do not identify the cause of individual errors or establish that more blood markers will reduce them.

The p-EASI studies provide the closest precedent for a usable serum-to-EASI calculation.[4–6] Their correlations and classification results encouraged molecular estimation; the present study adds explicit absolute errors, their distribution and within-person response checks in a local treatment cohort. Their use of serum alone differs from the present BSA–serum inputs. Measurement procedures can affect calibration, and both assay compatibility and case mix matter when moving a prediction rule between settings.[7, 8] The source-table local-application analyses therefore provide neither a fair superiority test nor evidence that population ethnicity explains a performance difference.

Both initial abrocitinib groups had lower adjusted itch scores at W2. This direction is consistent with the randomized JADE DARE comparison of abrocitinib 200 mg and dupilumab, which found more early itch responders with abrocitinib.[21] That trial used a PP-NRS decrease of at least four points, whereas our early target was an absolute PP-NRS of at most four; the rates are not directly comparable. Our W24 adjusted EASI and itch differences also persisted, so later improvement in all three groups should not be described as equivalence. This cohort provides observational comparisons, not a randomized head-to-head trial.

IgE and blood eosinophils require a different interpretation from skin severity. Changing follow-up composition explained why available-case IgE medians could rise while paired-patient estimates fell. The abro 200 mg direction remained sensitive to the missing-data model. Dupilumab trials have also reported transient blood eosinophil increases despite clinical efficacy.[22] That literature makes a simple monotonic-decline expectation inappropriate, but cannot reconstruct the unmeasured dupi W4 cell in this cohort.

Treatment context matters both for interpreting biomarkers and for comparing clinical outcomes. Patients receiving the higher initial abrocitinib dose had greater starting severity; subsequent reductions were selected after clinical response. Baseline adjustment and explicit maintenance pathways therefore address different questions. This distinction is consistent with randomized abrocitinib induction/withdrawal designs and observational dupilumab tapering studies, which define eligibility for maintenance or reduction before evaluating its success.[23, 24] The present results describe adjusted initial-strategy associations and observed later management, rather than the effect of assigning each patient an alternative dose. Differential missingness, endpoint-specific residual imbalance and response-selected follow-up limit these comparisons even after baseline adjustment. They also preserve itch and quality-of-life outcomes as complementary clinical domains; neither is adequately recovered by the same small serum–BSA panel.

The proposed management concept keeps the research EASI estimate alongside measured skin findings, symptoms, quality of life and safety (Fig. 1). It does not supply automatic escalation, reduction or stopping thresholds. Independent evaluation should test the frozen equation, assay procedure and a prespecified clinical decision together. Patient-level validation prevented a person's repeated visits from appearing in both training and testing, but the molecular panel had been selected earlier and week 24 had already been inspected in prior project analyses. The current evidence consequently supports within-cohort performance of the development procedure, with a clearly defined research formula for the next evaluation. Patient bootstrap intervals condition on saved predictions and omit model-development uncertainty; no independent validation of the single final formula has been performed. Follow-up overestimation and calibration slopes below one also argue against substituting estimates for clinical examination. Adequate numbers of active severe cases, independent clinical assessment and a prespecified tolerance for individual errors will determine whether the same simple calculation can support decisions across the full disease range.

Data and code availability

Analysis code, versioned parameters, aggregate source tables and figure sources are available at `https://github.com/yanax726/T2T`. Figures and tables can be rebuilt from aggregates; model refitting requires the declared restricted input snapshots. Patient-level clinical records, serum measurements, dates and free-text notes are not included in the public repository. Access conditions and the institutional contact for these restricted records require author confirmation.

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